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Where predictive information resides in endogenous A-to-I RNA editing

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Why particular RNA duplexes and adenosines undergo endogenous A-to-I editing remains incompletely understood. We used controlled prediction to examine substrate properties, tissue context, and local adenosine selection in human inverted-Alu RNA. Beyond the substantial signal in sequence and predicted structure, genomic organization predicted differences in editing burden between substrates, while a target-centered representation improved prediction at individual sites. The same genomic sites differed across tissues, yet the tested bulk contextual measurements did not consistently improve site prediction. Recovering additional valid labels and reducing the disproportionate influence of site-rich substrates improved learning. Most tested increases in supervision richness or model sophistication gave limited or adverse returns. A frozen combination of whole-duplex and local-augmented predictors improved performance across all five held-out benchmark endpoints. The search locates recoverable information at substrate and target scales while leaving tissue-associated variation unresolved by the tested contextual measurements. It motivates finer measurements matched to the cells and transcripts in which editing occurs, alongside further modeling.

Significance

A-to-I RNA editing helps cells distinguish their own RNA from viral RNA, yet why particular duplexes and adenosines are edited remains unclear. We separated this problem into whole-duplex properties, transcript and cellular context, and local adenosine selection. A predictive search across these scales found additional signal in genomic organization and correctly centered local context. Tissue-associated differences were visible, but the tested bulk RNA-binding protein, transcript, repeat, and chromatin measurements yielded no consistent predictive increment. Accounting for dependent sites and preserving target-local information helped more than the tested generic model extensions. This map identifies both useful information in current data and unresolved questions for measurements matched to the same cells and transcripts.

A-to-I RNA editing | ADAR | inverted Alu | RNA structure | machine learning

Adenosine deaminases acting on RNA (ADARs) convert adenosine to inosine in double-stranded RNA (dsRNA). The reaction diversifies transcripts and prevents endogenous duplexes from activating antiviral pathways (1–5). Most detected human editing occurs in paired Alu elements, whose abundance and orientation create imperfect intramolecular duplexes (6–10). Editing is selective at two levels: many potential duplexes show little editing, and neighboring adenosines in one predicted helix can have sharply different outcomes. The same genomic site can also differ among tissues (11, 12).

This selection problem has three biologically distinct scales. Whole-duplex sequence, folding, and genomic organization constrain whether a transcript can form a suitable substrate. Transcript processing, localization, regulator exposure, and enzyme state may determine whether that substrate is realized in a particular cell. Within an available duplex, nucleotide identity, mismatches, bulges, and local helix context influence which adenosine is edited (13–17). These questions require different measurements and controls. They are also distinct from a statistical issue in the benchmark: sites from one duplex are dependent observations, so site-rich substrates can exert disproportionate influence during learning.

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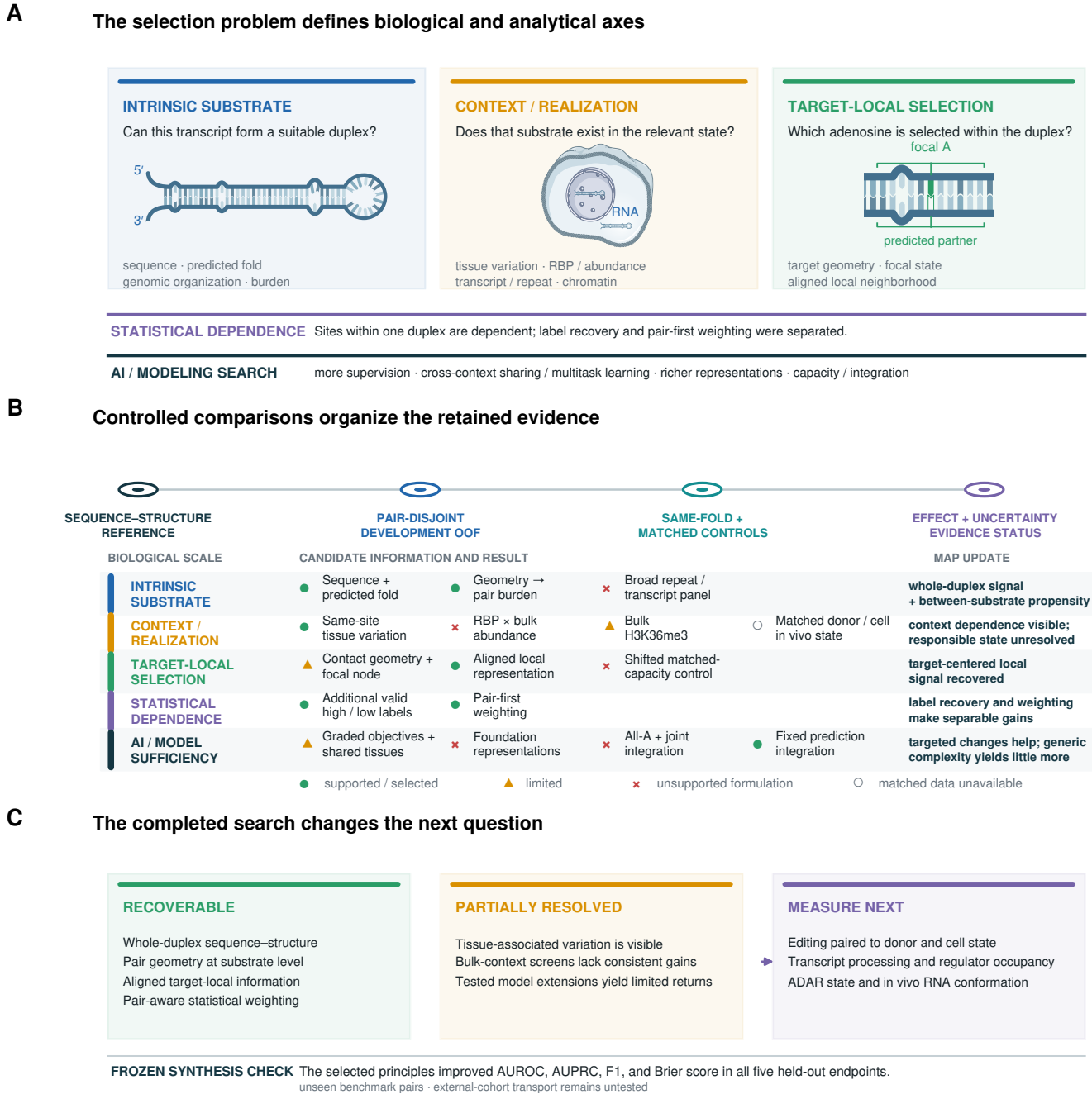


Fig. 1. Systematic search across the information space accessible to the benchmark. (A) Intrinsic whole-duplex suitability, transcript/cellular realization, and target-local selection define the biological axes; dependent observations and model sufficiency define separate analytical questions. Illustrations are conceptual. (B) Candidates were compared against a sequence–structure reference using pair-grouped development evaluation and relevant matched controls. Symbols distinguish supported/selected, limited, unsupported in the tested formulation, and deferred branches. Early residual screens were exploratory, not fully nested (Methods). (C) Recoverable signal and unresolved context motivate finer matched measurements. The frozen test evaluated only the selected synthesis on unseen benchmark pairs. SI Appendix, Table S10 records branch-level controls, evidence regimes, and dispositions.

This decomposition makes the unresolved problem experimentally tractable within a predictive benchmark. Adding a measurement or changing a representation asks whether it recovers signal unused by a fixed reference. The distinction matters: reorganizing the same sequence and structure can improve prediction without adding a new biological measurement. Null increments constrain the tested formulation; redundancy, noise, finite data, or model misspecification remain possible explanations. Matched comparisons and falsification controls can therefore probe recoverability without establishing mechanism.

Biological candidates were grouped by scale (Fig. 1). A parallel model-sufficiency axis examined richer supervision, cross-context sharing, pretrained representations, and greater capacity or integration. The search was adaptive: outcomes determined which branches advanced, stopped, or required unavailable measurements. The Methods and SI Appendix, Table S10 distinguish those dispositions and the evidence supporting them; only the frozen synthesis entered the final test.

AdarEdit established that sequence and predicted structure can predict site-level editing in this pair-partitioned benchmark (18). Here that model provides the common reference for a different question: which information scales account for the recoverable signal, what adds to it, where plausible measurements fail, and when added modeling sophistication yields diminishing returns. The search finds information at whole-substrate and target-local scales, detects tissue-associated variation that present bulk measurements do not resolve, and identifies pair-first weighting as a separate correction for dependent observations. The selected predictor is the final test of whether those supported principles combine on unseen benchmark pairs.

Results

Search design separates discovery from held-out evaluation. Figure 1 maps the questions and outcomes; SI Appendix, Tables S2 and S10 record the quantitative comparisons and branch dispositions. Unavailable donor-resolved state and in vivo conformation are deferred measurements, not negative results.

We analyzed 884 graph-buildable inverted-Alu substrates across four individual GTEx tissues and Combined, a pooled endpoint. Complete pairs were assigned without overlap to 566 training, 139 validation, and 179 test pairs. Discovery combined the training and validation pairs in pair-disjoint out-of-fold (OOF) comparisons with pair-clustered uncertainty. For final fitting, the published roles were restored: training fit the selected components, validation fixed checkpoints and thresholds, and the frozen specification was evaluated on the canonical test (Fig. 2). Development comparisons support branch-level interpretation; the held-out test evaluates only the final synthesis.

Whole-duplex sequence and structure define a strong but non-universal reference. Whole-duplex sequence and predicted structure already predict much of the benchmark's site ranking. The compact reference achieved within-context AUROC of 0.869–0.933 across the five human endpoints and 0.823–0.865 across three nonhuman datasets (Fig. 3A). These descriptive ranges span heterogeneous datasets and do not estimate a common effect.

Transfer was strongest among human endpoints. It varied more among nonhuman datasets, and human–nonhuman transfer was lower and asymmetric (Fig. 3B and C). Sequence–structure regularities extend across contexts without being universal. These datasets cannot separate the effects of lineage, substrate composition, assay, labeling, and cellular environment.

The reference can rank sites while leaving substrate availability, tissue state, and unmeasured conformations unresolved. Its AUROC is not a fraction of explained biological variance.

Genomic organization adds between-substrate information. The substrate-scale branch asked whether genomic organization contributes information absent from the folded graph. Separation between Alu arms correlated negatively with edited-site fraction in every individual tissue (Spearman $\rho = -0.334$ to -0.482; Fig. 4B). This distance is absent from the site-level input, which joins the arms with a fixed separator. Low-capacity models combining geometry with graph summaries predicted pair-level edited fraction with $R^2 = 0.050$–0.277 and outperformed graph summaries alone in every tissue (Fig. 4C; SI Appendix). Genomic organization therefore contributes reproducible between-substrate signal beyond the tested graph summaries. The modeled outcome is editing propensity; duplex realization and ADAR engagement were not measured.

Editing differs across tissues, but current bulk measurements do not resolve the variation. Editing outcomes varied among tissues even when genomic site identity was fixed. In a post-hoc analysis of development pairs, 1,221 of 74,758 sites observed in at least two individual tissues switched strictly between edited ($\geq 15\%$) and nonedited ($< 1\%$) states. Among 50,162 sites observed in all four tissues, 17.4% spanned at least 15 EditingIndex points. Alu-pair rankings on identical site sets still correlated across every tissue pair ($\rho = 0.727$–0.921 for mean EditingIndex; SI Appendix, Fig. S7). Stable substrate differences coexist with tissue-dependent site outcomes.

Predicted substrate burden did not translate reliably into better site-level classification. An outer-isolated gate added 0.0084 AUPRC and 0.0041 F1 over expanded row-wise training, but the F1 interval included zero and Brier score increased by 0.0051. A neural burden auxiliary tested against the stronger pair-first model also failed its prespecified criterion. Static geometry predicts pair-level propensity without identifying why individual sites depart from it.

Contextual-feature screens yielded no consistent residual increment at the site level. They tested replicated ENCODE eCLIP occupancy, sequence-based RNA-binding protein (RBP) predictions, tissue-average abundance and their interactions, with randomized or shuffled controls (19–22). Sequence-based predictions complemented eCLIP, for which Alu multimapping makes peak absence ambiguous. These early residual screens reused pooled first-stage OOF scores and are exploratory rather than fully nested (Methods). Separate pair-disjoint analyses of matched-site switches and continuous tissue differences also found no increment from the RBP proxies beyond tissue effects. The negative evidence concerns these measurements, not established RBP regulation (23–26).

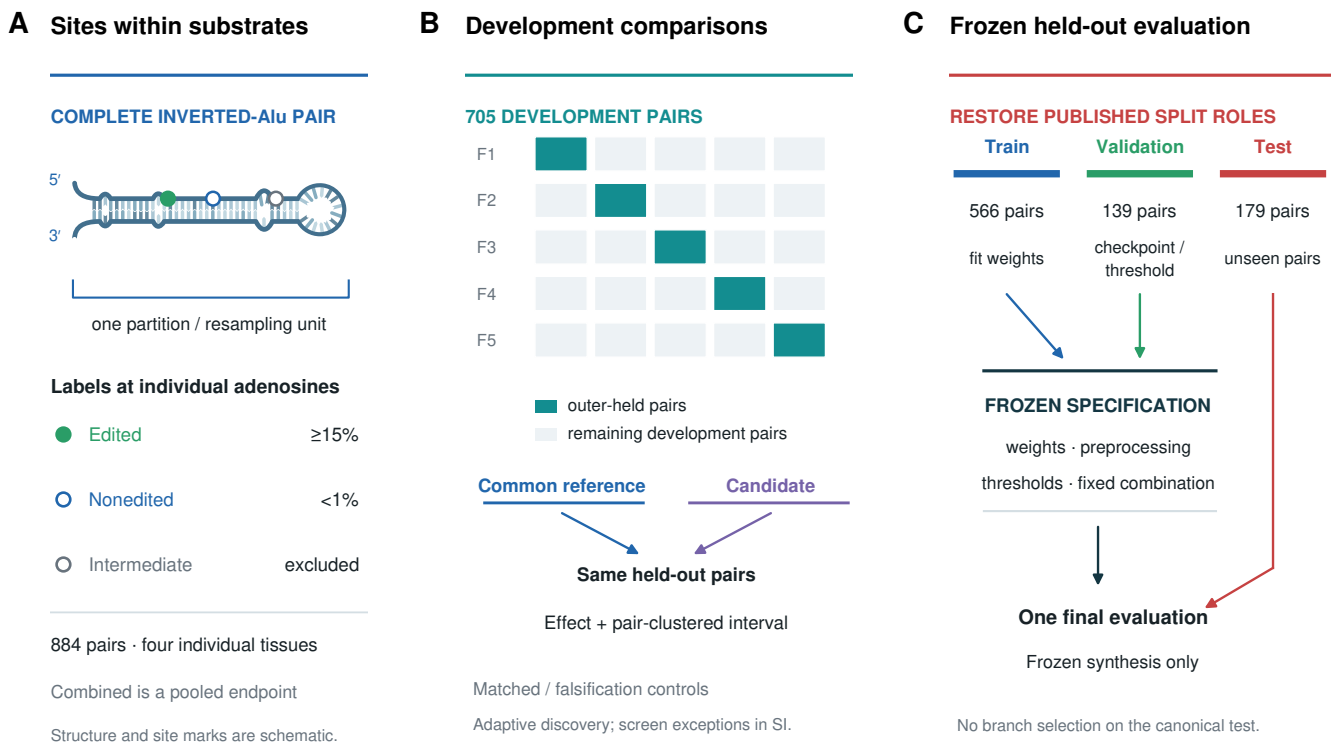


Fig. 2. Benchmark units and evaluation regimes. (A) Site labels are nested within 884 complete inverted-Alu pairs; structure and marked sites are schematic. Intermediate observations are excluded from the canonical task; Combined is pooled. (B) Five pair-disjoint outer folds define development comparisons. Neural candidate and reference models use the same folds and pair-clustered uncertainty; early pooled-score residual screens have the nestedness limitation described in Methods. Matched or falsification controls test specific attributions. (C) Final fitting restores the published split roles and freezes the specification before evaluating 179 unseen pairs. This test evaluates the synthesis, not each adaptively examined branch.

Transcript and repeat annotations likewise lacked a stable increment beyond the reference. Bulk H3K36me3 overlap yielded metric-specific effects, including a small liver AUPRC increment, without consistent site- or pair-level benefit (27, 28). Thus the available contextual measurements have not supplied a reliable site-level correction in the tested comparisons. They do not identify the source of the remaining error.

Pair-first weighting balances dependent observations across substrates. The statistical branch separated a sampling problem from the biological search. The canonical row set contains only part of the available high- and low-editing observations. Under uniform site sampling, a substrate's influence grows with its number of observations even though sites in one duplex share sequence, structure, genomic context, and often editing behavior.

A same-fold ablation separated label recovery from pair weighting in muscle (Fig. 5A; SI Appendix). Adding all high/low observations under row-wise sampling improved AUROC by +2.04 percentage points (95% pair-clustered interval +1.06 to +3.00) and F1 by +1.42 points (+0.21 to +2.57). Holding that pool and class balance fixed, then sampling a pair uniformly within each class before a site, contributed a further +2.27 AUROC points (+1.25 to +3.36) and +2.85 F1 points (+1.68 to +4.01). Label recovery and within-class substrate weighting made separable contributions.

The pair-first formulation improved the same-fold reference in all five endpoints (Fig. 5B); all 15 AUROC, AUPRC, and F1 intervals excluded zero. Pair-first sampling is therefore a weighting correction for dependent observations in this benchmark, not evidence for an additional biological determinant.

Pair-first learning improved log loss most for pairs with the fewest eligible sites, whereas the two largest site-count quintiles favored row-wise learning (SI Appendix, Fig. S8B).

Target-centered local context adds information beyond whole-duplex pooling. Whole-duplex pooling can obscure differences among candidate adenosines. Exploratory target-contact features improved ranking and Brier score but not F1. In a controlled neural comparison, explicitly retaining the focal graph node improved AUROC and AUPRC beyond an otherwise matched duplicate-global control. This localized signal unused by the pooled representation near the target.

The selected local-augmented graph predictor uses a 65-position representation centered jointly on the focal adenosine and its dot-bracket-aligned helix partner. A low-capacity dilated convolution encodes nucleotide identity, pairing state, focal and partner indicators, and signed relative position. Its output is fused with the whole-duplex graph embedding before classification. In muscle OOF data, the model reached AUROC/AUPRC/F1 of 0.9045/0.8936/0.8337, compared with 0.8885/0.8730/0.8208 for the pair-first whole-duplex-only predictor (Fig. 5C).

A matched-capacity control retained the same graph-plus-convolution architecture and optimization but shifted the local center by 96 nt. Performance fell to

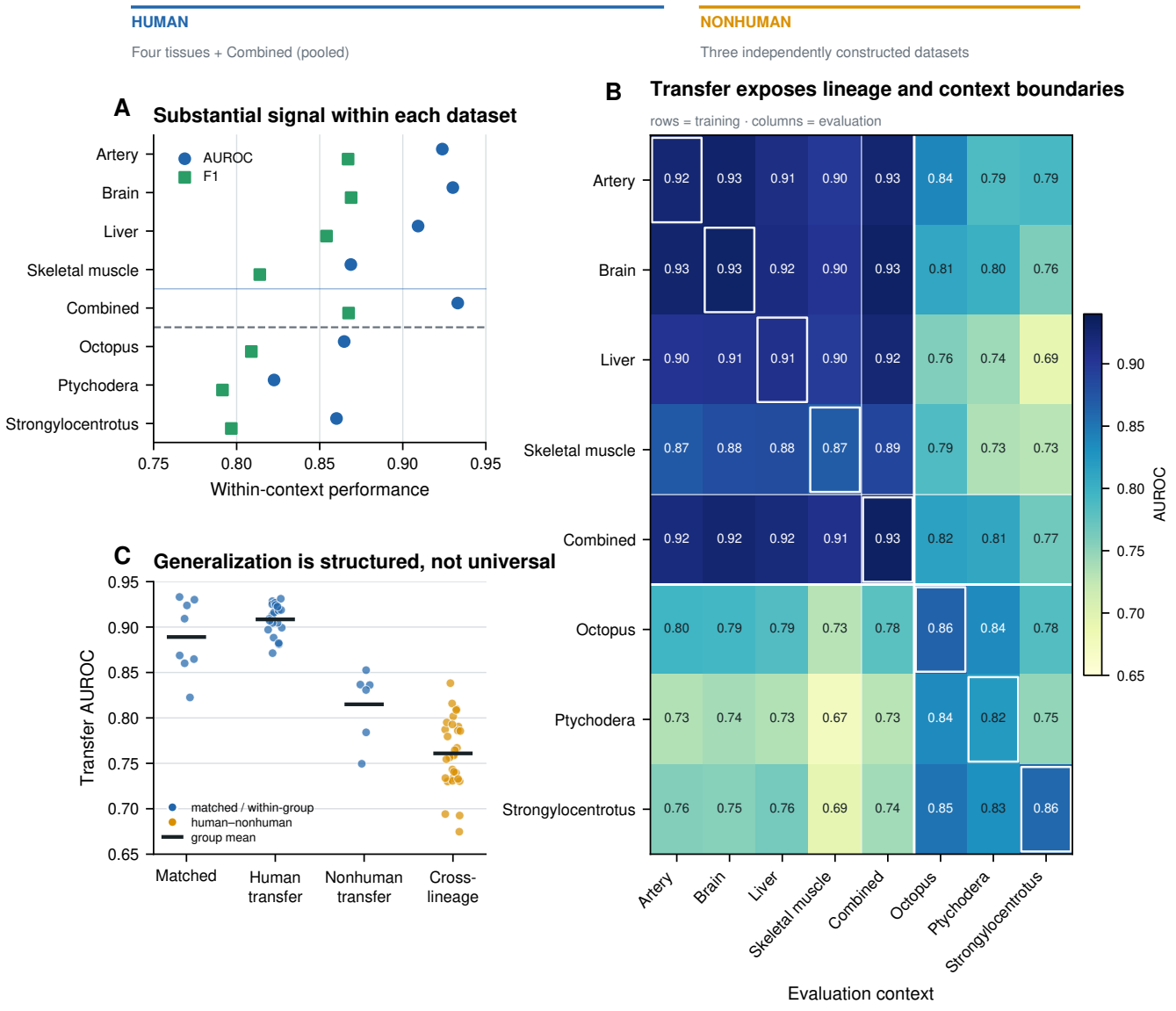


Fig. 3. Sequence and predicted structure carry substantial but non-universal signal. (A) Within-context AUROC and F1 for the retained compact sequence–structure baseline across five human endpoints and three nonhuman datasets. Combined is pooled. (B) AUROC when each training context is evaluated in every context; white lines separate human and nonhuman blocks, and outlined diagonal cells mark matched-context evaluations. (C) Transfer grouped by the relation between training and evaluation contexts; horizontal bars are group means. Dataset heterogeneity precludes causal attribution or meta-analysis.

0.7122/0.6943/0.7028. Aligned-minus-shifted differences were 0.1926 AUROC, 0.1993 AUPRC, and 0.1311 F1, with entirely positive pair-clustered intervals. The correctly centered, pairing-organized neighborhood is therefore important beyond added convolutional capacity. Shifting the full window does not isolate the predicted partner from that neighborhood. The aligned model's advantage over whole-duplex prediction also held in liver and in the secondary natural-prevalence analysis; the shifted-window control was tested in muscle.

Descriptive Brier improvements occurred for paired and unpaired targets and across every target-stem extent (SI Appendix, Fig. S8C and D). Gains were not confined to one of these structural strata.

Whole-duplex and target-local views remain complementary. The whole-duplex and local-augmented predictors made sufficiently different errors to benefit from combination. A fixed mean of logits improved the whole-duplex-only predictor by 0.0256/0.0308/0.0139 AUROC/AUPRC/F1 in muscle and 0.0182/0.0214/0.0179 in liver. All six pair-clustered intervals excluded zero (Fig. 5D).

The final predictor uses two independently trained pair-first components: a whole-duplex-only graph predictor and a graph predictor augmented with the aligned target/helix-partner convolution. Their logits are combined by a fixed 0.5/0.5 mean, with no weight search (Fig. 5E). This fixed combination retains both predictive views without fitting a flexible stacker to only hundreds of independent substrates.

In a post-hoc equal-pair analysis restricted to duplexes containing both labels, the fixed ensemble improved mean within-duplex AUROC over the pair-first component in muscle (+1.78 points; 95% pair-bootstrap interval +1.21

A

Evidence map: which information changes the recoverable signal?

INFORMATION LEVELS	Tested formulation	Outcome	Interpretation
	LEARNING & SUPERVISION		
LEARNING	Pair-first expanded supervision	● Selected	More labels; correct pair weight
	Continuous + ordinal auxiliary labels	▲ Limited	Small ranking gain
	Shared four-tissue training	▲ Limited	Transfer below preset criteria
	TARGET REPRESENTATION		
TARGET-LOCAL	Target/partner-aligned local view	● Selected	Passes shifted-window control
	Focal node / contact geometry*	▲ Limited	Target-position signal
	Frozen RNA-language embeddings	✖ Not supported	F1-only; no ranking gain
	Substrate-wide all-A decoder	✖ Not supported	Stopped for futility
	Supplied Bio-aware variant	▲ Limited	Features/capacity confounded
	Graph Transformer / GATv2 expansion	+ Not pursued	Not tested after local gains
	SUBSTRATE & CELLULAR CONTEXT		
SUBSTRATE / CONTEXT	Pair geometry / editing burden	◆ Supported†	Substrate propensity detectable
	Hierarchical engagement gate	▲ Limited	Ranking gain; unstable F1
	Bulk H3K36me3 overlap*	▲ Limited	Metric-specific screen effects
	RBP binding × bulk expression*	✖ Not supported	No consistent screen gain
	Transcript / repeat annotations*	✖ Not supported	No site-level gain
	Matched donor/cell multimodality	■ Deferred	Matched data unavailable
	Matched in vivo structure / isoforms	■ Deferred	Matched data unavailable
INTEGRATION	**INTEGRATION**		
	Fixed whole-duplex + local-augmented logits	● Selected	Complementary predictions
	Unified end-to-end architecture	✖ Not supported	Below frozen ensemble

*Pooled-score exploratory screens. Comparators and intervals: Table S2.

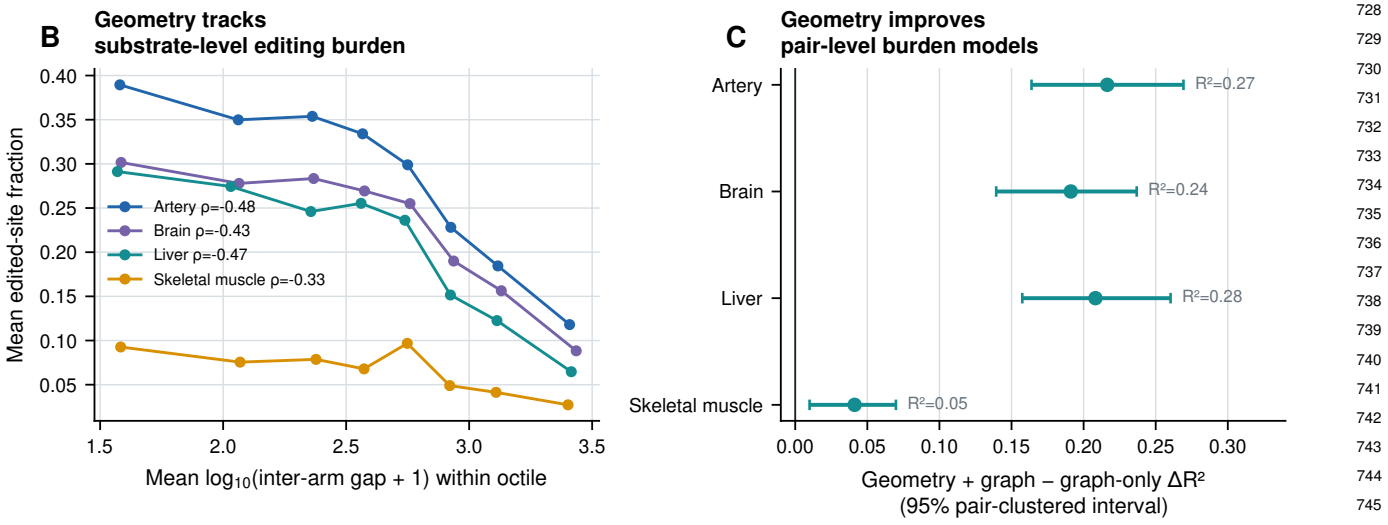


Fig. 4. The evidence map identifies a distinct substrate-level signal. (A) Search outcomes grouped by information scale. SI Appendix, Table S2 gives comparators, effects, and intervals. "Supported†" denotes an observation not yet useful for site prediction; "not supported" concerns the tested formulation. Asterisks mark pooled-score exploratory screens (Methods); within the focal/contact row, only the contact-feature screen has this limitation. Deferred measurements were unavailable. (B) Mean edited-site fraction across inter-arm-gap octiles in four tissues; ρ is calculated over individual development pairs. (C) Increment in pair-level R^2 after adding geometry to graph summaries. Bars are 95% pair-clustered intervals; labels give combined-model R^2.

to +2.40) and liver (+1.38 points; +1.06 to +1.70). The aligned-local component alone showed a clear within-duplex increment only in muscle, so this analysis supports complementary within-substrate ranking rather than a replicated local-component effect (SI Appendix, Fig. S8A).

Most tested model-expansion routes yield limited returns. The model-sufficiency search asked whether four common routes to greater modeling power could close the residual gap: richer supervision, cross-context or multitask learning, richer or pretrained representations, and greater capacity or integration. Comparisons within each class used a compatible reference. Expanding the pool of valid high/low labels helped, whereas binary–ordinal–continuous supervision added only a small ranking effect and reduced natural-prevalence performance when combined. Shared four-tissue learning produced limited

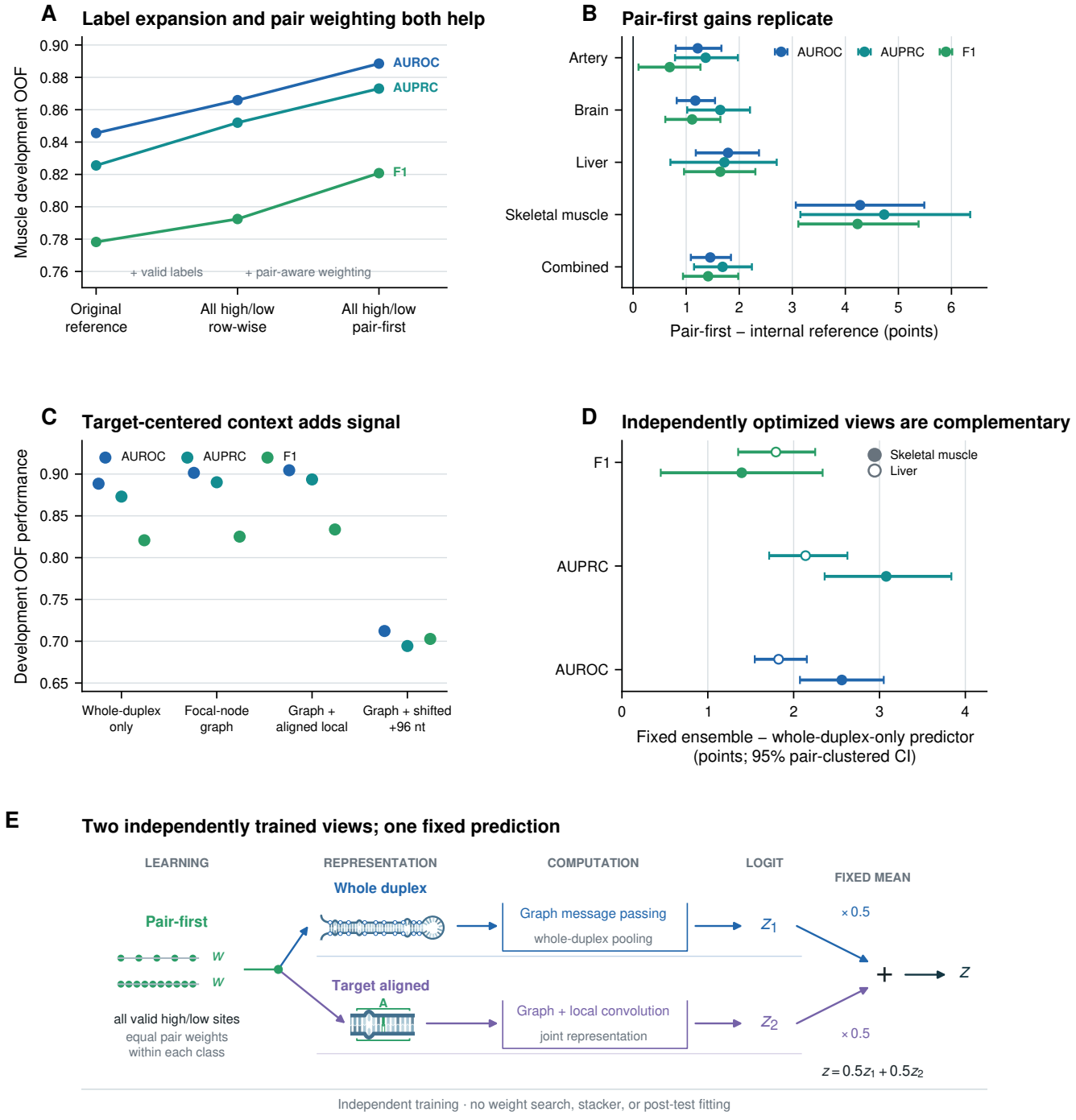


Fig. 5. Pair-first learning and target-centered representation provide complementary gains. (A) Same-fold ablations separate recovery of additional valid high/low labels from pair-first weighting. (B) Pair-first expanded supervision improves the internal reference across all five endpoints; points and bars are pair-clustered medians and 95% intervals. (C) Whole-duplex pooling, focal-state retention, and aligned-local fusion form an evidence ladder; an otherwise identical local model is shifted by 96 nt. (D) A fixed equal-logit ensemble improves its whole-duplex-only component in muscle and liver. (E) The predictor combines independently trained whole-duplex-only and graph-plus-local models without searched weights or post-test fitting. The RNA illustrations are conceptual. All panels show development evidence.

transfer and worse mean Brier score (Fig. 4A; SI Appendix).

Representation changes were selective. Focal-node retention and aligned local encoding improved target prediction; frozen RiNALMo embeddings did not add ranking signal beyond matched controls (29). Substrate-wide decoding and the preregistered unified end-to-end model underperformed their references, while burden-based hierarchical formulations remained limited. The supplied Bio-aware variant changed both features and architecture, so it did not isolate capacity; a separate graph-transformer expansion was not pursued. Among completed controlled tests, substrate-aware weighting and target-centered representation supplied clearer gains than the tested generic extensions. This comparison does not define a ceiling for future architectures.

A frozen synthesis improves prediction on unseen benchmark pairs. The final test asked whether principles selected across the search combine on unseen benchmark pairs. All components, checkpoints, thresholds, weights, code, and hashes were frozen before canonical held-out evaluation. The selected predictor improved AUROC, AUPRC, and F1 in all five reported endpoints (Fig. 6A), with mean gains of 0.0277, 0.0261, and 0.0253 over reproduced supplied-checkpoint AdarEdit. Every endpoint had a positive 95% pair-clustered interval for all three metrics, and Brier score decreased throughout (Fig. 6B and C; SI Appendix, Table S1).

Combined overlaps the individual tissue endpoints, so the mean is descriptive rather than a five-tissue meta-analysis. The frozen combination generalizes to unseen pairs within this benchmark. Its component-level attribution rests on the controlled development comparisons above.

At frozen validation-selected thresholds, the selected predictor corrected 32.0–50.6% of baseline false positives while converting 1.7–5.9% of baseline true positives into false negatives (Fig. 6D). A secondary matched-recall analysis is reported in SI Appendix, Table S9.

Discussion

The search separates two ways to recover signal from endogenous A-to-I editing. Genomic organization supplies a measurement absent from the folded input; target-centered encoding makes better use of sequence and pairing information already available. Both improve prediction at their respective scales. Matched sites also differ across tissues, but the tested bulk and cell-line proxies have not provided a reliable site-level correction. Retaining the limited and unsupported branches distinguishes these findings from a list of successful features: the map records where plausible additions did, and did not, improve the available comparisons.

The substrate result extends earlier observations that distance between oppositely oriented Alus predicts aggregate Alu editing (16). Within a common predictive design, geometry explained pair-level burden beyond graph-derived sequence and structure summaries, but did not become a reliable site-level correction. At the nucleotide scale, prior biochemical and comparative work implicated sequence preference, mismatches, bulges, and helix context (13–15, 17). The aligned-versus-shifted comparison now shows that correctly centered, pairing-organized local context contributes beyond whole-duplex pooling. It does not isolate the predicted partner or establish a molecular mechanism. Pair-first weighting solves a different problem by preventing substrates with more observed sites from dominating estimation.

The context branch identifies a measurement hypothesis, not an observed latent state. Perturbation studies implicate RBPs and chromatin-associated regulation in editing (23–27). Here, tissue-associated differences coexist with limited predictive value from cell-line occupancy, tissue-average abundance, transcript and repeat annotations, and bulk chromatin. These findings motivate measuring regulator exposure and transcript state at the same biological resolution as editing. They do not establish that coarse measurement is the sole reason the tested proxies underperformed.

The secondary modeling search provides a complementary constraint. Graded supervision and cross-tissue sharing gave limited gains; frozen RNA foundation representations, substrate-wide decoding, and connected end-to-end integration did not outperform their respective references on the selection criteria. The strongest improvements came from class-balanced pair-first learning and correctly centered local encoding. Generic model sophistication alone did not close the gap in the tested regime. Residual error could still reflect finite data, model misspecification, measurement noise, structural uncertainty, unmeasured transcript or cellular state, or biological stochasticity. The present data cannot apportion these sources.

The benchmark covers graph-buildable human inverted-Alu substrates and tissue summaries. Predicted structure is a surrogate for in vivo conformations; eCLIP is cell-line-specific and repeat-censored. Nonhuman transfer confounds lineage, substrate class, assay, and labeling. The canonical labels are balanced and exclude intermediate observations, so natural-prevalence analyses remain secondary. Component attribution comes from adaptively selected, nonmultiplicity-adjusted development comparisons; early pooled-score residual screens additionally require fully nested replication. Tissue comparisons share substrates and are not independent cohorts. The frozen test assesses unseen pairs within the benchmark construction, not transportability to an independently generated human cohort (18).

The frozen two-view predictor checks whether the selected principles work together: they improved held-out discrimination, classification, and probability accuracy. The next biological test is to measure editing and its candidate regulators on the same substrates in the same donors or cells. Transcript processing, ADAR isoform and interferon state, localization, regulator occupancy, quantitative chromatin, and in vivo RNA conformation are motivated candidates, not a proven sufficient measurement set. Such data would test which context variables add to the substrate- and target-scale information recoverable here.

Materials and Methods

Search design and evidence classification. The adaptive search grouped candidates by intrinsic substrate properties, transcript/cellular context, and target-local selection, with separate axes for dependent observations and model sufficiency. The latter covered richer supervision, cross-context sharing, richer/pretrained representations, and capacity or integration. Priorities reflected biological rationale, information beyond the reference, measurement compatibility, leakage risk, and effective sample size. Development branches used pair-grouped comparisons against a compatible reference, with randomized, shifted-location, or matched-capacity controls where relevant. Selected branches met recorded promotion criteria and entered the synthesis; supported findings were retained without selection; limited effects were metric- or context-specific. Not supported denotes failure of the tested formulation. Deferred branches lacked suitable data or a validated resource; superseded branches were not pursued after stronger evidence emerged. These adaptive comparisons are not multiplicity-adjusted confirmation, and the coverage taxonomy is not a claim that the entire program was preregistered. SI Appendix, Tables S2 and S10 distinguish controlled comparisons, exploratory screens, descriptive characterization, and untested branches. Only the frozen synthesis entered the final canonical evaluation.

Benchmark and labels. We used the AdarEdit v2 reproducibility release at upstream revision `5c8b57d70cdc2eb1c5e168431013275bcc9ed0c5`. The benchmark contains 884 graph-buildable inverted-Alu pairs. Human sites were labeled edited at EditingIndex $\geq 15\%$, nonedited at EditingIndex

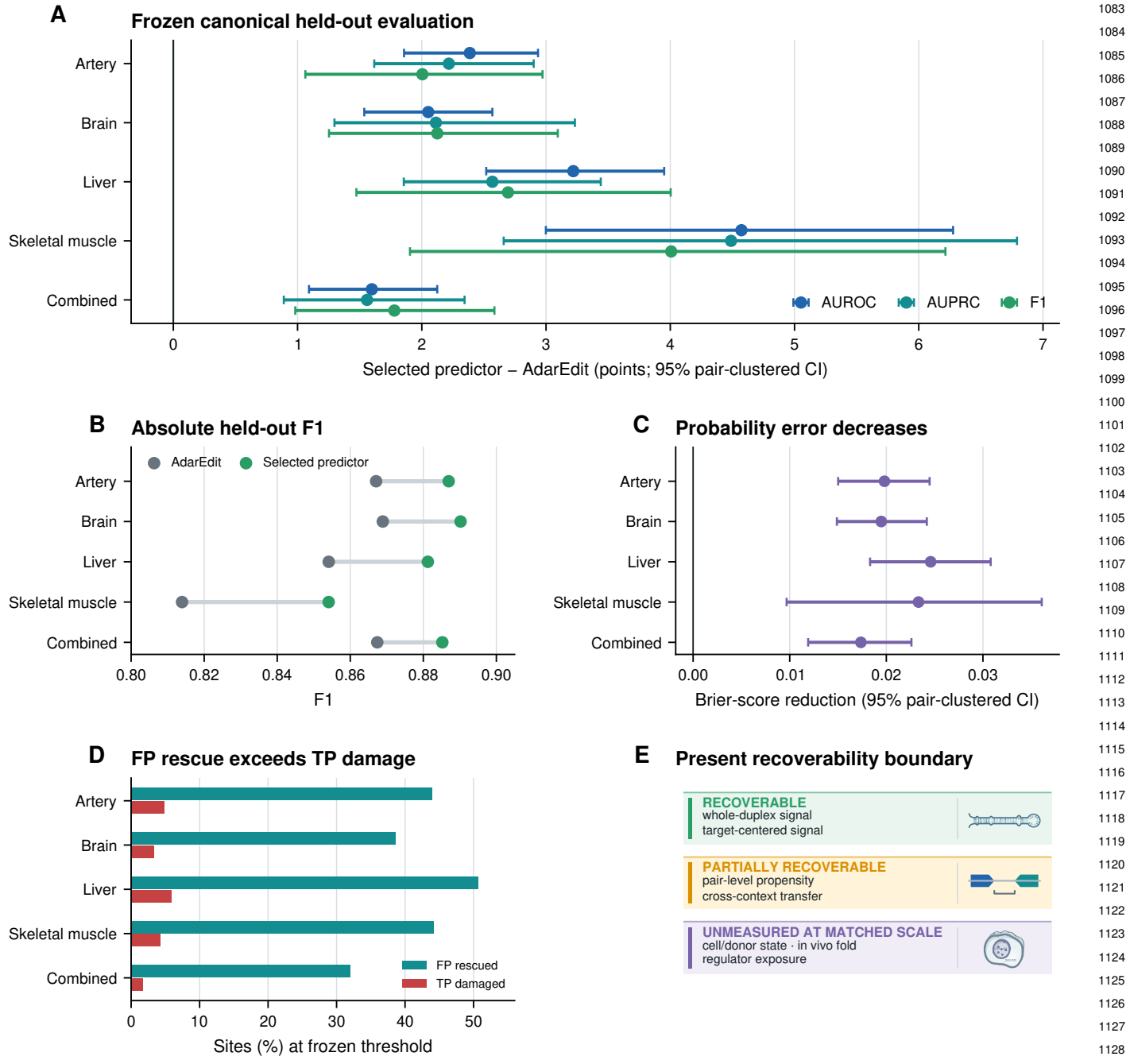


Fig. 6. Frozen evaluation confirms within-benchmark generalization and summarizes what current measurements recover. (A) Selected predictor minus reproduced supplied-checkpoint AdarEdit for AUROC, AUPRC, and F1; bars are 95% pair-clustered bootstrap intervals. (B) Absolute F1. (C) Reduction in Brier score. (D) Baseline false positives corrected and baseline true positives converted to false negatives at frozen validation-selected thresholds. (E) Information captured, partially resolved, and unavailable at matched biological scale; tier-specific icons are conceptual. Combined is pooled. All quantitative panels use the frozen canonical held-out evaluation.

$< 1\%$, and intermediate otherwise. The canonical task excludes intermediate observations. We verified pair identities and the fixed 566/139/179 train/validation/test assignment before analysis. Combined was used only as its published pooled endpoint.

Development protocol. Discovery used the original train and validation pairs in 5 outer folds by pair ID. Neural comparisons excluded the outer fold from training, preprocessing, checkpoint choice, and threshold choice; one other fold provided internal validation. Early residual screens instead fitted pair-disjoint second-stage models to pooled first-stage OOF logits. Their meta-training scores could depend on the current outer fold through first-stage fitting, so they are exploratory, not fully nested estimates (SI Appendix, section A). Later outer-isolated gates and foundation fusion used fold-specific first-stage fits. Supplied checkpoints were reproduced using aggregate test metrics before discovery; candidate development did not access the canonical test.

Models. The first component retained the compact AdarEdit graph attention encoder. Each epoch allocated equal high/low class counts, sampled pairs uniformly among those eligible for each class, and then sampled a site within each selected pair. The canonical-row update budget was unchanged. The second component was trained independently with the same graph encoder. It added a 65-position target/helix-partner-aligned representation encoded by two dilated convolutional layers, then fused local and graph embeddings. The selected probability was $\sigma[(\mathrm{logit}\, p_{\mathrm{whole}} + \mathrm{logit}\, p_{\mathrm{augmented}})/2]$, with weights fixed at 0.5/0.5 and no weight search. The shifted-window control preserved architecture and optimization while

moving the local center 96 nt. SI Appendix gives the feature definitions and remaining protocols.

Statistics and final evaluation. Metrics were AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, MCC, and Brier score. Thresholds maximized validation F1 within each run. Pair-clustered bootstraps resampled Alu pairs with all their sites: development comparisons used 1,000 or 2,000 replicates, the post-hoc within-duplex analysis used 5,000, and the final comparison used 2,000. These intervals condition on the fitted models and do not include variability from rerunning training or adaptive selection. Code, inputs, weights, epochs, thresholds, and checkpoint hashes were frozen before the selected predictor's held-out evaluation; no subsequent model selection or tuning occurred. Natural-prevalence and matched-recall analyses are secondary (SI Appendix).

AI-assisted preparation. OpenAI Codex assisted with manuscript editing, source-code review, and figure-rendering code. RNA and cell illustrations were adapted from credited DBCLS/TogoTV and Servier Medical Art SVGs (SI Appendix, Artwork credits). Earlier GPT Image 1.5 illustrations are excluded. Author confirmation of the complete tool/model-version disclosure and of journal eligibility for AI-assisted diagram code is pending.

Data, Materials, and Software Availability. Code, experiment configurations, registries, source tables, figure scripts, retained derived artifacts, frozen checkpoints, and provenance are available at https://github.com/Scientific-Computing-Lab/ADARPro. Third-party downloads and deterministic matrices omitted from GitHub are listed with source URLs, roles, checksums, and regeneration instructions in the data manifest. The AdarEdit benchmark is available from https://github.com/Scientific-Computing-Lab/AdarEdit. No new human participants were recruited and no new primary sequencing data were generated.

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